Review

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The Evolving Blueprint: A Survey on Automated Optimization of LLM-Based Multi-Agent Systems

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Abstract

The rapid rise of Large Language Model (LLM) agents is driving a fundamental paradigm shift in Multi-Agent Systems (MAS) research, moving from manually orchestrated static architectures toward automated configuration and optimization. Despite its significant potential, this frontier lacks a systematic and rigorous survey with clearly defined operational boundaries. To address this gap, this paper provides a comprehensive review of Automated MAS Optimization, formally anchoring it as the P4 paradigm within a six-stage evolutionary framework spanning from Foundation LLMs (P0) to Agentic Swarms (P5). We introduce precise mathematical definitions for core concepts, establishing a unified MAS configuration space that encompasses agent-level, system-level, and underlying components, and formulate the optimization objective as a holistic system-utility maximization problem. Furthermore, we partition P4 into three operationally distinct sub-paradigms based on the orthogonal dimensions of optimization timing and effect persistence: Design-Time Adaptive MAS, Test-Time Adaptive MAS, and Self-Evolving MAS. Guided by this taxonomy, we systematically review over 200 state-of-the-art works, covering both general methodologies and domain-specific applications. Beyond algorithmic perspectives, we critically examine key supporting issues including benchmarking, evaluation, and safety, while analyzing the evolutionary trajectory toward decentralized, emergent P5 Agentic Swarms. Finally, we identify core open challenges and propose future research directions centered on holistic configuration co-optimization, life-cycle evaluation, endogenous safety mechanisms, and the controllable transition from P4 to P5. This survey aims to provide a rigorous theoretical foundation and strategic navigation for researchers and practitioners in this rapidly evolving field.

Keywords: large language models; multi-agent systems; automated configuration optimization; multi-agent collaboration; self-evolving architectures; agentic swarms

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1. Introduction

LLM-based agent systems are rapidly evolving from single-agent problem solving toward more complex forms of multi-agent collaboration [1–3]. Early research mainly focused on equipping a single agent with tool use, memory, planning, and reflection, enabling it to exhibit a degree of autonomy on complex tasks [4–6]. As task complexity increased, however, a growing body of work shifted toward multi-agent systems (MAS), where multiple agents with differentiated responsibilities collaborate to solve tasks that a single agent cannot reliably handle alone [7–9]. This shift has significantly expanded the capability frontier of agentic systems, while also making system performance increasingly dependent on system-level design choices, such as how agents are specialized, how they communicate, how they are scheduled, and how intermediate results are integrated [1,10,11].

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Early LLM-based MAS largely relied on manual workflows and collaboration structures, resulting in high development costs, limited scalability, and weak adaptability to task feedback [12,13]. Against this backdrop, the research frontier is shifting toward automated MAS optimization, namely the automatic generation, configuration, and optimization of multi-agent collaboration. This direction targets the overall MAS configuration θ to seek optimal system setups for a task distribution under budget, safety, and privacy constraints. From a capability perspective, it enables systematic search across topology, roles, prompts, memory, tools, and scheduling, overcoming manual design limits while better adapting to complex environments [11]. From an engineering perspective, it reduces the costs of building, debugging, and deploying systems while improving scalability and generalization. Recently, automated MAS optimization has rapidly evolved from scattered exploratory efforts into a highly active research area with a rich methodological structure [14]. As shown in Figure 1, its recent development exhibits strong cumulative growth, distinct methodological lineages, and broad visibility across major venues. Consequently, it is no longer a loose collection of refinements but an independent research direction demanding dedicated and systematic treatment.

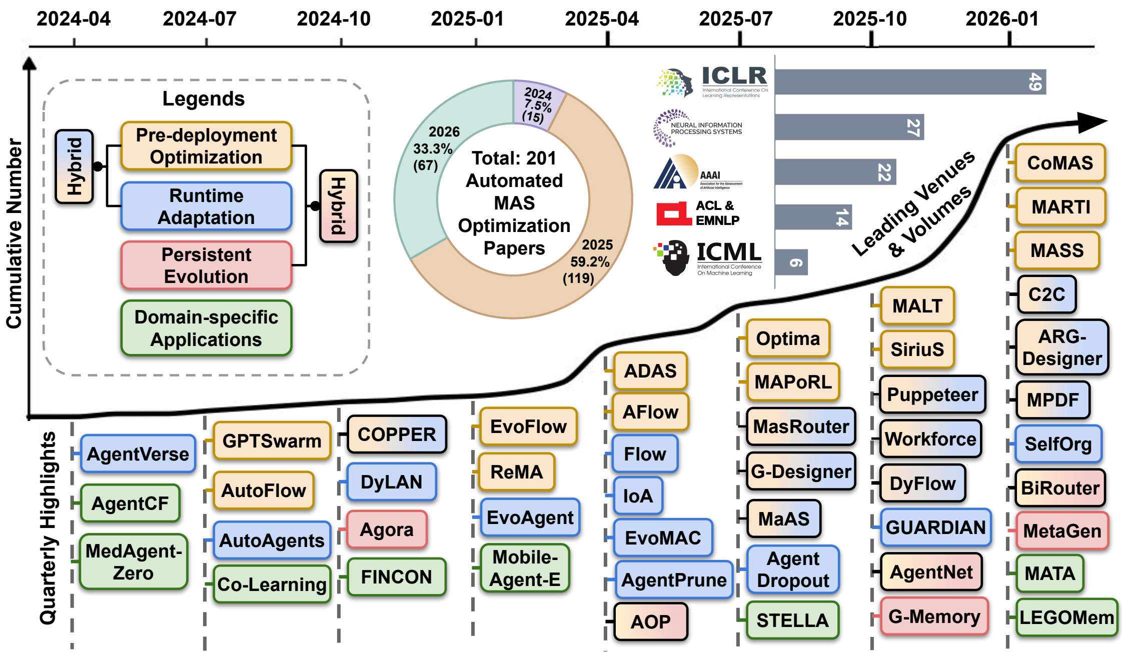

Figure 1. The rapid emergence of automated MAS optimization. This figure illustrates the rapid crystallization of this research area since early 2024. The main timeline tracks cumulative publication growth, highlighting representative works quarterly and categorizing them by major methodological paradigms. Inset panels detail the annual publication distribution and leading academic venues.

To study this emerging area systematically, we first place it within an evolutionary taxonomy of agentic systems progressing through *P0 (Phase 0) Foundation LLM*, *P1 Static Single Agent*, *P2 Automated Single Agent*, *and P3 Orchestrated MAS*, ultimately reaching *P4 Automated MAS* and *P5 Agentic Swarm*. Within this spectrum, we explicitly position P4 as the central focus. We then formalize this paradigm as the optimization of a comprehensive system configuration space. This target transcends individual agent capabilities to encompass collaborative dimensions including topology, roles, prompts, memory, tools, scheduling, and communication protocols. Building upon this foundation, we categorize existing methods through a systematic framework governed by two fundamental questions of automation: the timing of optimization and the persistence of its effects. Finally, we examine critical aspects including evaluation, benchmarking, and safety, thereby establishing a rigorous analytical lens and clear boundaries for future research.

By applying this evolutionary taxonomy, we compare the most influential and closely related surveys emerging from the explosion of LLM-based agent and MAS research in Table 1. As shown, existing literature primarily focuses on single agents or orchestrated systems from P1 to P3. While some surveys include P4 in their scope [14,17,18,21], their limited literature coverage and lack of operational boundaries restrict them to broad conceptual narratives, making it difficult to systematically characterize this emerging paradigm. To bridge this critical gap and establish a structured understanding of the field, this survey provides the first dedicated and in-depth review centered exclusively on automated MAS optimization.

Table 1. Comparison with representative related surveys, with emphasis on their coverage of Automated MAS (P4). FD: Formal Definition; OB: Operational Boundary.

Survey	Date	Primary Scope[†]	Focus of Survey		Automated MAS (P4)		
			Primary	Secondary	# Papers	FD	OB[‡]
Guo *et al.* [1]	Jan 2024	MAS	P3	—	3	✗	○
Wang and Ma *et al.* [15]	Mar 2024	Autonomous agents	P1, P2	P3	1	✗	○
Li *et al.* [7]	Oct 2024	MAS	P3	P1	1	✗	○
Xi *et al.* [4]	Jan 2025	Agents	P1, P2	P3, P5	1	✗	○
Tran *et al.* [8]	Jan 2025	MAS collaboration	P3	—	5	✗	○
Luo *et al.* [5]	Mar 2025	Agents	P1, P2	P3	3	✗	○
Ferrag *et al.* [11]	Apr 2025	Autonomous agents	P0, P1, P2	P3	0	✗	○
Wang and Pan *et al.* [16]	Jun 2025	LM-based agents	P3	P1, P2	2	✗	○
Ye *et al.* [17]	Aug 2025	Collective intelligence	**P4**, P5	P3	12	✗	◐
Fang *et al.* [18]	Aug 2025	Self-evolving agents	P2, **P4**	P3	29	✓	◐
Gao *et al.* [14]	Jan 2026	Self-evolving agents	P2, **P4**	—	15	✓	◐
Du *et al.* [19]	Feb 2026	Agent optimization	P2	P4	11	✗	◔
Zhu *et al.* [20]	Feb 2026	MAS	P3	P4	16	✗	◐
Xiang *et al.* [21]	Feb 2026	Self-evolving agents	P2, **P4**	—	21	✓	◐
This Survey	Apr 2026	Automated MAS optimization	**P4**	P5	**201**	✓	●

[†] Except for Wang and Pan *et al.* [16], "agents" and "MAS" in the Primary Scope denote *LLM-based* systems. [‡] ○: Not Covered; ◔: Descriptive; ◐: Developing (substantial but not yet fully reusable); ●: Operational (explicit, reusable, and decision-ready).

Our main contributions are summarized as follows:

- We presents a dedicated survey of automated MAS optimization as an independent research direction. By situating it within the evolutionary framework of agentic systems from P0 to P5, we propose a formal definition, an operational boundary, and a configuration-centric analytical lens, thereby clearly delineating the scope and theoretical foundation of this field.
- Driven by optimization timing and effect persistence, we introduce an organizational framework to systematically categorize a substantial volume of relevant literature. By covering both the primary methodological spectrum and domain-specific applications, this survey forms a comprehensive characterization of automated MAS optimization.
- Extending beyond methodological and applied summaries, we discuss critical supporting dimensions including evaluation, benchmarking, and safety. Furthermore, we contextualize automated MAS within the broader evolutionary chain progressing toward P5 agentic swarms to identify key future research directions for the field.

The remainder of this paper is organized as follows. Section 2 establishes the formal foundation of automated MAS, detailing its position within the evolutionary framework from P0 to P5, its optimization configuration space, and its operational boundaries. Sections 3 through 5 systematically review the literature across three core sub-paradigms of automated MAS optimization: design-time adaptive MAS, test-time adaptive MAS, and self-evolving MAS. Section 6 summarizes domain-specific instantiations shaped by real-world constraints. Section 7 discusses early explorations and the broader evolutionary trajectory toward agentic swarms. Finally, Sections 8 and 9 analyze benchmarking and safety challenges, before Section 10 concludes with future research directions.

2. Formal Foundations of Automated MAS: Taxonomy and Optimization

Prior work lacks dedicated survey coverage and rigorous, operationalizable definitions for automated multi-agent systems (MAS) as a distinct paradigm. Existing literature often conflates terms like "self-adaptive," "self-evolving," and "autonomous," thereby complicating systematic methodological comparisons. To bridge this gap, this section introduces a formal framework, illustrated through-

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out the panels of Figure 2, that first situates MAS development within a structured six-paradigm taxonomy while partitioning automated MAS into three distinct sub-paradigms, second provides precise mathematical definitions for core concepts and the configuration-centric optimization problem, and third establishes operational boundary conditions that unambiguously isolate automated MAS from adjacent paradigms. Stricter than prior descriptive treatments, these formalisms constitute an independent technical contribution of this survey.

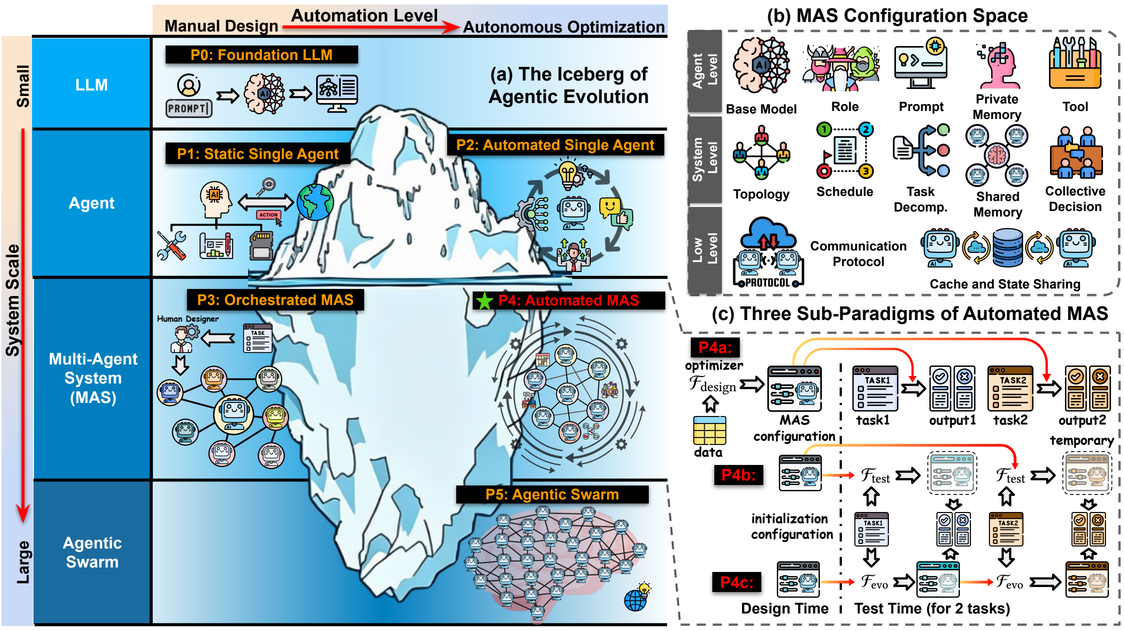

Figure 2. Formal foundations of automated MAS: a formal framework encompassing (a) **evolutionary taxonomy:** situating MAS development within a structured six-paradigm progression from foundation models to agentic swarms, with automated MAS (P4) serving as the central focus; (b) **configuration space:** a multi-level decomposition of the system into agent-level, system-level, and low-level components; (c) **optimization sub-paradigms:** partitioning automated MAS into design-time adaptive (P4a), test-time adaptive (P4b), and self-evolving (P4c) categories based on optimization timing and effect persistence.

2.1. Evolution of LLM-Based Agentic Systems

The transition from foundational language models to large-scale autonomous MAS is categorized into six paradigms from P0 to P5, differentiated by system scale and automation level. As illustrated by the iceberg metaphor in Figure 2 (a), foundation models and single agents from P0 to P2 constitute the visible tip of the agentic landscape, while the profound systemic complexity resides primarily beneath the surface. This survey focuses on the submerged mass of **P4 automated MAS**, a research frontier situated at the intersection of multi-agent collaboration and autonomous configuration optimization. This paradigm serves as the central focal point before the field transitions toward the emergent coordination of P5 agentic swarms.

- *P0 – Foundation LLM.* The pre-trained large language model serves as the sole computational unit without external tools or planning loops. Capabilities are governed by fixed parameters, with prompt engineering being the only intervention.
- *P1 – Static Single Agent.* Integration of tool use, memory, and planning yields an agent with a sense-plan-act cycle. The configuration is manually designed and remains static post-deployment.
- *P2 – Automated Single Agent.* The configuration of a single agent is autonomously determined or refined by an optimization mechanism rather than manual intervention.
- *P3 – Orchestrated MAS.* Multiple agents interact under a specific topology to tackle complex tasks. However, the system architecture relies on manual specification or human-in-the-loop refinement, lacking an autonomous optimization process.

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- ★ *P4 – Automated MAS.* This paradigm employs an automated optimization process to determine or improve MAS configurations through systematic search, evaluation, and updates without human intervention in the optimization loop. A more detailed definition can be found in Section 2.3.
- • *P5 – Agentic Swarm.* The frontier moves beyond bounded configuration optimization to enable decentralized and emergent coordination among large agent populations.

2.2. Formal Definitions of Core Concepts

We now establish the formal definitions that underpin the remainder of this survey. A MAS comprises two or more autonomous agents interacting within a shared environment. Unlike single-agent systems, MAS introduces system-level organizational structures, most notably the topology Γ, that coordinate agent connectivity and unlock capabilities beyond individual limits through role specialization, parallel execution, and cross-verification.

2.2.1. Basic Formulation

Definition 1 (Environment)**.** *An environment $\mathcal{E}$ is a black-box system mapping an execution trajectory τ and task goal g to a feedback signal:*

$$r = \mathcal{E}(\tau, g), \quad r \in \mathbb{R} \cup \mathcal{L},$$

where r encompasses both scalar rewards and natural-language evaluations. Unlike traditional POMDPs, this formulation directly utilizes linguistic feedback to drive optimization.

Definition 2 (Agent)**.** *An agent α_i is a four-tuple (ϕ_i, P_i, M_i, W_i) representing the base model, prompt, memory, and tool set respectively. The agent policy π_i at time t is:*

$$\pi_i(\cdot \mid o_t, P_i, M_i) \in \Delta(\mathcal{A}_i \cup W_i).$$

Definition 3 (Multi-Agent System)**.** *A MAS Π is defined as $(\Gamma, \{\alpha_i\}_{i=1}^{n})$ for $n \geq 2$. The topology $\Gamma = (\mathcal{V}, \mathcal{C})$ specifies the directed communication channels and routing rules between agent nodes.*

Definition 4 (Task)**.** *A task $\mathcal{T} = (\mathcal{E}, g)$ produces a trajectory τ and feedback r under system Π. This feedback serves as the primary signal for subsequent optimization and adaptation processes.*

2.2.2. MAS Configuration Space

Definition 5 (MAS Configuration Space)**.** *The configuration space of a MAS is defined as:*

$$\Theta = \Theta_{\text{high}} \times \Theta_{\text{low}},$$

where the high-level configuration space Θ_{high} captures the semantic structure of the MAS, delineating agent identity, function, and organizational logic to serve as the primary target of MAS optimization:

$$\begin{aligned}\Theta_{\text{high}} = \underbrace{\Gamma}_{\text{topology}} \times \underbrace{\{\phi_i\}}_{\text{models}} \times \underbrace{\{R_i\}}_{\text{roles}} \times \underbrace{\{P_i\}}_{\text{prompts}} \times \underbrace{\{M_i\}}_{\text{private memory}} \times \underbrace{\mathcal{M}_{\text{shared}}}_{\text{shared memory}} \\ \times \underbrace{\{W_i\}}_{\text{tools}} \times \underbrace{\Lambda}_{\text{scheduling policy}} \times \underbrace{\mathcal{D}}_{\text{task decomposition}} \times \underbrace{\mathcal{J}}_{\text{collective decision}},\end{aligned}$$

and the low-level configuration space Θ_{low} captures the communication infrastructure, governing how information physically flows and is shared across the system:

$$\Theta_{\text{low}} = \underbrace{\mathcal{C}}_{\text{communication protocol}} \times \underbrace{\mathcal{K}}_{\text{cache/state}}.$$

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As illustrated in Figure 2 (b), the components of Θ are grouped by functional scope into three categories, where the first two together form the high-level configuration space. Critically, the system-level and low-level categories are distinctive to MAS relative to single-agent systems, introducing the complex coordination and communication challenges that constitute the primary focus of this survey.
Agent-level components. These operate independently for each agent instance α_i.

- ϕ_i: **Base models.** The language model instance assigned to each agent, determining its underlying reasoning and perception capabilities.
- R_i and P_i: **Role configurations and prompts.** These two components jointly determine agent identity and behavior. R_i specifies the functional identity within the system, such as a planner, executor, critic, or verifier. P_i specifies how that role is instantiated behaviorally through instructions and task descriptions. They constitute distinct optimization dimensions since the same role R_i may be instantiated with varying prompts P_i, and automated prompt optimization modifies P_i without altering the underlying R_i.
- M_i: **Private memory.** The per-agent memory state, including short-term context history and long-term knowledge, accessible exclusively to agent α_i.
- W_i: **Tool sets.** The callable APIs, functions, and external services defining the agent's action interface toward the external environment.

System-level organizational components. These govern global collaboration dynamics.

- Γ: **Topology.** The potential connectivity structure among agents, specifying permissible communication channels. Γ defines the structural space of possible connections independent of whether those connections are activated during a specific execution.
- Λ: **Scheduling policy.** The operational execution policy governing agent activation order, routing rules, and dynamic instantiation conditions. Whereas Γ specifies which interactions are structurally possible, Λ determines which are realized at runtime and in what sequence.
- $\mathcal{D}$: **Task decomposition.** The semantic breakdown of the global task goal g into sub-goals and their subsequent assignment to agents. $\mathcal{D}$ dictates what sub-task each agent pursues, whereas Λ governs when and how that sub-task is executed.
- $\mathcal{J}$: **Collective decision mechanism.** The aggregation and arbitration rules by which multiple agent outputs are resolved into a final collective judgment, such as peer-review protocols, multi-agent debate, and majority voting. Different judgment rules induce distinct collective behaviors even under identical topologies and task decompositions.
- $\mathcal{M}_{\text{shared}}$: **Shared memory.** The cross-agent public memory and coordination infrastructure. Unlike private memory M_i, $\mathcal{M}_{\text{shared}}$ is jointly accessible across agents and encompasses storage substrates, retrieval policies, update rules, and retention mechanisms.

Low-level components. These manage foundational data exchange.

- $\mathcal{C}$: **Communication protocol.** Message format specifications, synchronization paradigms including synchronous or event-driven rules, and semantic compression applied to inter-agent messages. Optimization over $\mathcal{C}$ falls within scope when it directly affects the quality, fidelity, or structure of inter-agent information flow.
- $\mathcal{K}$: **Cache and state sharing.** Cross-agent key-value cache reuse and intermediate state management. This dimension is in scope when it alters the context accessible to agents or the semantics of knowledge sharing, but falls outside our scope when serving purely as an engineering optimization for inference efficiency without affecting decision-making.

2.2.3. MAS Optimization

Definition 6 (Utility Function)**.** *A utility function $U : \Theta \times \mathcal{T} \to \mathbb{R}$ maps a system configuration θ and a given task $\mathcal{T}$ to a scalar performance value. This value is derived from the environmental feedback r obtained by executing $\mathcal{T}$ under configuration θ.*

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Definition 7 (MAS Optimization)**.** *MAS optimization seeks an optimal configuration θ^* within a feasible set $\mathcal{B} \subseteq \Theta$. Operational constraints including token limits, latency boundaries, cost budgets, safety protocols, and privacy requirements uniquely induce this feasible set. The formal objective is to maximize the expected utility across a specific task distribution $p(\mathcal{T})$:*

$$\theta^* = \arg\max_{\theta \in \mathcal{B}} \mathbb{E}_{\mathcal{T} \sim p(\mathcal{T})}\left[U(\theta, \mathcal{T})\right].$$

Crucially, the explicit optimization target is the holistic system configuration rather than the isolated capability of any individual agent.

Exhaustively searching the feasible space $\mathcal{B}$ is generally computationally intractable. Consequently, practical implementations typically bypass global optimization to target a strategically chosen subset of components within Θ.

2.3. Automated MAS and Its Sub-Paradigms

Building on the foregoing definitions, we now formally define Automated MAS (P4), the primary focus of this survey. Prior work has introduced related notions [14,18,21], but these are often stated at a descriptive level and do not provide sufficiently sharp operational boundaries for systematic classification. The definitions below therefore adopt a stricter and more fine-grained formulation, which constitutes an independent technical contribution of this survey.

Definition 8 (Automated MAS)**.** *A system is an* ***Automated MAS*** *(P4) if it satisfies both:*

- ***(M)*** ***Multi-agent condition:*** *The system comprises $n \geq 2$ agents organized under a topology Γ.*
- ***(A)*** ***Automation condition:*** *There exists a systematic optimization process $\mathcal{F}$ that determines or improves MAS configuration θ through search, evaluation, selection, update, or generation, without per-iteration human intervention in the optimization loop.*

Human practitioners retain the architectural authority to specify an initial configuration θ_0, delimit a search space $\Theta_0 \subseteq \Theta$, formulate the optimizer $\mathcal{F}$, and define the operational constraints that induce the feasible set $\mathcal{B}$. However, the optimization of the MAS configuration must proceed without per-iteration human intervention, regardless of when or how $\mathcal{F}$ is invoked.

As illustrated in Figure 2 (c), we partition automated MAS into three sub-paradigms based on two orthogonal dimensions: optimization timing and effect persistence. Timing dictates whether the configuration mechanism is solidified prior to deployment or dynamically triggered at runtime. Persistence determines whether runtime adaptations remain transient to the current query or manifest as durable modifications influencing subsequent tasks. These resulting sub-paradigms, designated P4a through P4c, are not mutually exclusive. A single system can seamlessly integrate multiple automated optimization forms across different stages of its operational lifecycle.

Definition 9 (Design-Time Adaptive MAS)**.** *An automated MAS is designated as* ***Design-Time Adaptive MAS*** *(P4a) if the optimization of its configuration mechanism is completed exclusively prior to deployment. Formally, a design-time optimizer $\mathcal{F}_{\text{design}}$ leverages historical information $\mathcal{H}$ and a task distribution $p(\mathcal{T})$ to produce:*

$$\omega = \mathcal{F}_{\text{design}}(\mathcal{H}, p(\mathcal{T})),$$

where ω denotes either a fixed global configuration θ^* or a frozen configuration policy π_ψ. The deployed policy π_ψ may directly map an input query q to a query-specific configuration $\theta(q)$ or be invoked to determine task-conditional configurations during runtime.

The optimizer $\mathcal{F}_{\text{design}}$ can materialize as various computational mechanisms, including search algorithms such as Monte Carlo Tree Search [13] and evolutionary search [22], gradient-based optimizers [23,24], and LLM-as-optimizer schemes [12]. A human-provided initial configuration θ_0 may serve as a warm start without *invalidating the P4a classification. Consequently, P4a characterizes systems where the configuration mechanism itself is mathematically derived through pre-deployment optimization, regardless of whether the resulting frozen artifact operates in a static or query-conditioned manner at runtime.*

Definition 10 (Test-Time Adaptive MAS)**.** *An automated MAS is designated as **Test-Time Adaptive MAS** (P4b) if it invokes an automated process $\mathcal{F}_{\text{test}}$ during deployment to dynamically determine, instantiate, or optimize a task-specific configuration and execution strategy for a given query q:*

$$(\theta_q, a_q^*) = \mathcal{F}_{\text{test}}(\omega, q, \mathcal{E}),$$

where ω denotes the deployed configuration artifact available at runtime, θ_q denotes the task-specific configuration explicitly generated for query q, and a_q^* denotes the resulting output or execution trajectory. Unlike P4a, the automated optimization process actively executes at runtime in direct response to the incoming query. Unlike P4c, any resulting adaptation remains strictly transient. The underlying global configuration θ remains structurally rigid, ensuring that no optimization effects persist beyond the resolution of the immediate query.

Definition 11 (Self-Evolving MAS)**.** *An automated MAS is designated as **Self-Evolving MAS** (P4c) if it employs an active evolution engine $\mathcal{F}_{\text{evo}}$ at runtime to persistently modify its global configuration based on accumulated execution experience:*

$$(\theta_{t+1}, \Theta_{t+1}) = \mathcal{F}_{\text{evo}}(\theta_t, \Theta_t, \tau_t, r_t), \qquad \theta_t \in \Theta_t,\ \theta_{t+1} \in \Theta_{t+1},\ \Theta_{t+1} \supseteq \Theta_t.$$

In stark contrast to P4a and P4b, the configuration undergoes durable modification directly through ongoing task execution. This evolution materializes either as systematic value updates within an existing configuration space or as an intrinsic expansion of the adjustable component space itself. To qualify under this paradigm, the update process must rigorously satisfy three criteria:

- ***Persistence.*** *The resulting modifications must persist beyond the immediate query lifecycle to fundamentally influence system behavior on subsequent tasks.*
- ***Experience-driven optimization.*** *The evolutionary trajectory is strictly propelled by the system's empirical execution experience interacting with the environment, rather than following any externally prescribed schedule.*
- ***Configuration-level restructuring.*** *The runtime evolution must intrinsically rewrite the deployed baseline MAS configuration itself, distinguishing it from superficial optimizations that merely adjust transient outputs for a current query.*

Along the persistence axis, these sub-paradigms consolidate into two broader categories. P4a and P4b collectively constitute **self-adaptive MAS**, as their optimization processes yield no durable configuration modifications across sequential tasks. Conversely, P4c uniquely represents **self-evolving MAS**, defined by its capacity to accumulate persistent structural changes through ongoing execution experience. While the persistence boundary strictly divides these categories, the timing dimension permits non-exclusive hybrid architectures, such as integrating design-time P4a optimization with runtime P4b or P4c mechanisms. By formalizing this strict persistence boundary, we resolve the prevalent conceptual conflation in prior literature [14,18,25]. Empirically, this analytical lens reveals that self-adaptive methods dominate the current research landscape, whereas genuinely self-evolving systems remain scarce. Ultimately, this rigorous classification provides an operational criterion for organizing the literature and solidifies the evolutionary narrative bridging automated optimization toward the emergent complexities of P5.

2.4. Boundary Conditions Across Paradigms

Having established the mathematical foundations for P4 and its constituent sub-paradigms, we now delineate the operational boundary conditions that unambiguously isolate automated MAS from adjacent paradigms, accompanied by a systematic comparative analysis. Such definitional precision is imperative. Existing literature frequently suffers from conceptual ambiguity, leading to the inconsistent and contradictory classification of fundamentally similar systems.

2.4.1. Boundary Between P2 and P4

The primary demarcation between P2 and P4 is governed by the explicit optimization target: whether the automated process refines an isolated agent or a holistic multi-agent system. A nuanced edge case emerges when a multi-agent setup is employed exclusively during the training phase. In such scenarios, the decisive criterion remains the focal point of the optimization rather than the mere presence of multiple interacting entities. If the multi-agent environment functions purely as a training substrate to enhance a single agent whose parameters are updated in isolation, the paradigm strictly remains P2. This classification holds because the final deployed artifact is a standalone agent, and the auxiliary multi-agent structure is discarded post-training, a methodology exemplified by systems including ATM [26] and CORY [27].

2.4.2. Boundary Between P3 and P4

The decisive boundary between P3 and P4 lies in whether the system employs a systematic optimization process $\mathcal{F}$ that operates autonomously without per-iteration human intervention. While both paradigms can exhibit dynamic, task-specific behaviors without explicit human prompting, not all such dynamics qualify as automated optimization. A prevalent source of ambiguity arises from P3 systems that dynamically adjust agent selection or routing for a current task purely through underlying LLM reasoning, a pattern exemplified by systems including KAMAC [28] and LbMAS [29]. To rigorously resolve such boundary cases, we formalize two distinct types of system dynamics.

Definition 12 (Type-A and Type-B MAS Dynamics)**.** ***(i) Type-A Conditional Inference.*** *The system adjusts its behavior through one-shot LLM reasoning over the current task without invoking a systematic optimization engine $\mathcal{F}$. It merely executes a heuristic sufficiency judgment rather than engaging in rigorous search, evaluation, or iterative updates toward an optimum.* ***(ii) Type-B Automated Optimization.*** *The system rigorously satisfies automation condition A outlined in Definition 8.*

Type-A dynamics characterize solely the dynamic subset of P3, whereas a substantial portion of P3 systems remain strictly static with human-hardcoded topologies, roles, and prompts. By contrast, all P4 systems intrinsically satisfy the foundational automation condition and therefore exclusively exhibit Type-B dynamics.

2.4.3. Boundary Between P4 and P5

Definition 13 (Agentic Swarm)**.** *A system transitions into an* ***agentic swarm*** *P5 regime when its hallmark collective capabilities are characterized by three intrinsic properties:*

- ***Emergent global behavior.*** *The system's distinctive capabilities arise primarily from large-scale local interactions across the agent population and cannot be adequately reduced to a bounded, explicitly designed coordination structure or a finite-dimensional MAS configuration $\theta \in \Theta$.*
- ***Decentralized population-level coordination.*** *Coordination is dominated by distributed interaction dynamics across an extensive agent population rather than by the centrally optimized or directly attributable collaboration structures typical of P4.*
- ***Regime-dependent emergence.*** *The system's hallmark capabilities become qualitatively visible only beyond a critical population scale or interaction density, representing a phase transition rather than a mere quantitative improvement as agent count increases.*

The scale-dependent nature of these systems fundamentally drives their emergent behavior. Because a P5 system's collective capabilities manifest through population-level interaction dynamics, they inherently resist reduction to any bounded, configuration-level description. This irreducibility

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constitutes the definitive boundary separating P5 from P4. P4 systems reside within the regime of bounded, optimization-ready architectures. Their behaviors remain primarily attributable to identifiable configuration components, such as topology, scheduling, or task decomposition, and admit a finite control description. This structural bounding holds true even for self-evolving MAS under P4c; although such systems persistently modify their architectures over time, these modifications remain strictly at the configuration level rather than emerging in the swarm sense. Consequently, scaling up a P4 system may yield quantitative performance gains but does not fundamentally alter its operational paradigm. By contrast, once a system satisfies the emergent criteria in Definition 13, it crosses into the P5 regime, where collective capability is governed by emergence rather than bounded by configuration. While mature P5 systems do not yet exist, pioneering efforts including PIANO [30] have begun to systematically probe this uncharted frontier.

3. Design-Time Adaptive Multi-Agent Systems

Design-Time Adaptive Multi-Agent Systems (P4a), the largest and most mature category in current Automated MAS research, optimize configurations primarily before deployment. This offline optimization reduces inference-time latency, costs, and complexity while enhancing stability, resulting in a fixed configuration or a frozen generation mechanism used during inference. However, this design-time focus inherently limits continuous runtime adaptation; adaptiveness reflects the offline training distribution rather than ongoing evolution based on post-deployment feedback. Consequently, while P4a automates optimization without human intervention, it operates within a bounded, design-time regime, with methods categorized into **offline model training**, **offline search**, and **offline policy training** based on their primary design-time optimization efforts.

3.1. Design-Time Adaptive MAS via Offline Model Training

Offline model training constitutes the largest and most practically established, yet conceptually concentrated, subgroup within P4a by directly optimizing agent or backbone LLM parameters at design time for fixed runtime deployment. Because this line of work predominantly targets model parameters while largely omitting higher-level MAS configurations, the resulting methods share broadly similar optimization pipelines. Consequently, their primary novelty lies less in introducing fundamentally new configuration mechanisms than in addressing specific multi-agent coordination bottlenecks. Given this procedural homogeneity, we organize these methods into six distinct categories based on their targeted coordination objectives rather than low-level training details, as summarized in Table 2 and illustrated below, thereby clearly differentiating approaches driven by fundamentally distinct coordination challenges.

- **Learning explicit collaboration and role specialization:** Rather than relying on emergent coordination, these methods optimize collaboration directly. For example, MAPoRL [35] uses Multi-Agent Reinforcement Learning (MARL) to co-train LLM agents for strategic multi-turn interactions instead of mere prompt-based simulation.
- **Improving learning signal quality and training stability:** To prevent training collapse and lazy-agent behaviors, these methods refine credit assignment across agents and steps. Dr. MAMR [54] achieves this via causal influence estimation and normalization debiasing to yield reliable optimization signals.
- **Improving data coverage, sample efficiency, and exploration:** Focused on acquiring more informative training experiences, methods like TEE [61] maximize trajectory entropy to encourage diverse cooperative exploration.
- **Improving generalization across tasks, scales, and structures:** To transfer policies beyond their original settings, methods like HiSSD [69] extract common and task-specific skills from offline data, adapting to unseen tasks with varying agent populations.

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- **Improving robustness and safety under realistic conditions:** Targeting reliable behavior under constraints or distributional shifts, methods like MOSDT [79] combine policy self-distillation with safety-aware representations to jointly optimize returns and safety.
- **Improving communication efficiency and interaction quality:** Treating communication as a direct optimization target rather than a byproduct, methods like Optima [87] train the MAS to jointly balance task performance, token efficiency, and readability.

A small subset of offline model training explicitly incorporates non-model MAS components, typically as auxiliary targets. For instance, CC-MADDPG [86] and PARCO [43] couple policy learning with communication optimization, JoyAgents-R1 [57] jointly evolves models and private memory, Agent0 [66] integrates tool use, and HCPO [44] augments training with a conductor-level policy. Conversely, a few methods centralize communication protocol optimization. ExpoComm [77] combines model training with exponential topologies, memory-based processors, and grounding objectives for scalable communication. GLC [88] learns a discrete, compressed, and grounded protocol via joint policy learning, autoencoding, language grounding, and contrastive alignment. Though exceptional, these demonstrate that model-centered offline training can occasionally extend to other MAS configuration components.

Table 2. Categorizing P4a offline model training methods by primary coordination bottleneck/objective.

Primary Bottleneck / Objective	Methods
Learning Explicit Collaboration and Role Specialization	CMAT [23], ReMA [31], Multiagent finetuning [32], ACC-Collab [33], MARFT-A [34], MAPoRL [35], MALT [36], SiriuS [37], AT-GRPO [38], CoMAS [39], AgentPO [40], MAGRPO [41], AgentCDM [42], PARCO (+$\mathcal{C}$) [43], HCPO (+Λ) [44]
Improving Learning Signal Quality and Training Stability	COALA-PG [45], MHGPO [46], M-GRPO [47], HyperMARL [48], QCoFr [49], RAC [50], CT-MARL [51], AT-GRPO [38], CoMAS [39], MARTI [52], MARSHAL [53], Dr. MAMR [54], GPVD [55], Dr. MAS [56], JoyAgents-R1 (+M_i) [57]
Improving Data Coverage, Sample Efficiency, and Exploration	DITS [58], DR-AC [59], INS [60], TEE [61], TRPE [62], S2Q [63], CEE [64], AgentEvolver (+$\mathcal{M}_{\text{shared}}$) [65] , Agent0 (+$W_i$) [66], HCPO (+$\Lambda$) [44]
Improving Generalization Across Tasks, Scales, and Structures	FlickerFusion [67], DoF [68], HiSSD [69], BiKT [70], SUBSAMPLE-MFQ [71], MA-MPL [72], STAIRS-Former [73], MAC-Flow [74], DiGS-CP [75], ADAPT [76], PARCO (+$\mathcal{C}$) [43], ExpoComm (+$\mathcal{C}$) [77]
Improving Robustness and Safety Under Realistic Conditions	ComaDICE [78], MOSDT [79], DSID-POMDP [80], MORNAVI [81], EPI [82], BATPAL [83], DrIGM [84], WCMASAC [85], CC-MADDPG (+$\mathcal{C}$) [86]
Improving Communication Efficiency and Interaction Quality	DiGS-CP [75], Optima (+$\mathcal{C}$) [87], CC-MADDPG (+$\mathcal{C}$) [86], ExpoComm (+$\mathcal{C}$) [77], GLC (+$\mathcal{C}$) [88]

[†] Methods marked with (+·) denote additional configuration components, beyond the base model component ϕ_i (in particular, model parameters here), that are also explicitly optimized by the corresponding methods.

3.2. *Design-Time Adaptive MAS via Offline Search*

Unlike training-based P4a methods, this group directly targets MAS configuration via design-time offline search. Methodologically similar to test-time optimization (P4b), they search before deployment, yielding a lighter inference footprint and avoiding per-query test-time overhead, though the offline search remains expensive [117]. Consequently, generalization is typically task-level rather than query-level. Table 3 characterizes these methods by optimization configuration space, paradigm, artifact form, and secondary objectives, further organizing them into agent- and system-level categories based on their primary optimization targets. Overall, this group is still dominated by system-level methods, most of which take topology as the central search target, while typically not explicitly optimizing the low-level communication layer.

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Table 3. Summarizing P4a methods using offline search through search-centric characterization.

Level	Method†	Search-Centric Characterization: Space / Target	Paradigm	Artifact Form	Objective Beyond Utility	Hybrid P4 Type
Agent Level	AgentSquare [89]	$P_i \times M_i \times W_i \times \mathcal{D}$	LLM-as-optimizer	Modular Agent	Module Synergy	✗
	OMAC [90]	$\Gamma \times P_i \times \Lambda$	LLM-as-optimizer	Prompt Set	—	✗
	AgentInit [91]	$R_i \times \mathcal{D}$	LLM-as-optimizer	Agent Team	Composition	✗
	Artemis [92]	$P_i \times W_i$	Evolutionary	Agent Configuration	Cost-Performance Tradeoff	✗
	TUMIX-Evolve [93]	$R_i \times P_i \times W_i$	LLM-as-optimizer	Agent Set	Coverage-Performance Tradeoff	✗
	MAEL [94]	M_i	Memory-based	Experience Pool	Cross-Task Experiential Learning	✗
System Level	ADAS [12]	$\Gamma \times P_i \times \mathcal{D}$	LLM-as-optimizer	Code	—	✗
	AFlow [13]	$\Gamma \times P_i$	Search-based	Code	—	✗
	Yuksel *et al.* [95]	$\Gamma \times R_i \times \Lambda \times \mathcal{D}$	LLM-as-optimizer	Code	—	✗
	Nexus Architect [96]	$\Gamma \times P_i \times W_i$	LLM-as-optimizer	Code	Generalization	✗
	A²Flow [97]	$R_i \times P_i$	LLM-as-optimizer	Code	Generalization	✗
		$\Gamma \times P_i$	Search-based	Code	—	✗
	JudgeFlow [98]	$\Gamma \times P_i$	LLM-as-optimizer	Code	—	✗
	SCALE [99]	$\Gamma \times P_i$	LLM-as-optimizer	Code	Cost-Performance Tradeoff	✗
	AutoFlow-C [100]	$\Gamma \times P_i \times \Lambda \times \mathcal{D}$	LLM-as-optimizer	NL Program	—	✗
	SwarmAgentic [101]	$\Gamma \times R_i \times \Lambda \times \mathcal{D}$	Search-based	NL Program	—	✗
	DebFlow [102]	$\Gamma \times P_i \times W_i$	Debate-based	Workflow	—	✗
	MermaidFlow [103]	$\Gamma \times P_i$	Evolutionary	Flowchart	Executability	✗
	AutoFlowGen [104]	$\Gamma \times \mathcal{D}$	LLM-as-optimizer	DAG	—	✗
	Cognify [105]	$\Gamma \times \phi_i \times P_i \times \mathcal{D}$	Search-based	Directed Graph	Quality-Cost-Latency Tradeoff	✗
	SOHM-D [106]	Γ	RL-based	Link Distribution	—	✗
	AgentXRay [107]	$\Gamma \times R_i \times W_i$	Search-based	Workflow	Workflow Reconstruction	✗
	EvoFlow [22]	$\Gamma \times \phi_i \times P_i$	Evolutionary	Workflow Library	Cost-Performance Tradeoff	P4b
	ANN [108]	$\Gamma \times R_i \times P_i \times \mathcal{D} \times \mathcal{J}$	LLM-as-optimizer	Layered Graph	—	P4b
	AgentDropout [109]	Γ	RL-based	DAG	Cost-Performance Tradeoff	P4b
	MetaAgent [110]	$\Gamma \times R_i \times P_i \times W_i$	LLM-as-optimizer	FSM	Robustness	P4b
	ReSo [111]	$\Gamma \times \phi_i \times R_i \times \mathcal{D}$	Search-based	Agent Graph	—	P4b
	AgentNet [112]	$\Gamma \times M_i \times \Lambda$	Search-based	DAG	Decentralization	P4c
	MATTRL [113]	$\mathcal{M}_{\text{shared}}$	Memory-based	Experience Pool	Distribution-Shift Robustness	P4b
	GPTSwarm [114]	Γ	RL-based	DAG	Workflow Orchestration	✗
		P_i	LLM-as-optimizer	Prompt	—	✗
	H-Swarms [115]	Γ	Search-based	DAG	Collaborative Generation	✗
		ϕ_i	Search-based	Model Weights	Individual Contribution	✗
	MASS [116]	P_i	LLM-as-optimizer	Prompt	—	✗
		$\Gamma \times W_i$	Search-based	Workflow	—	✗

[†] Multi-row methods indicate that different configuration components are optimized through distinct search pipelines. Hybrid P4 types shaded in gray.

At a finer granularity, agent-level offline search optimizes individual agents or local combinations rather than system topology. LLM-based methods optimize distinct components via generation, revision, or selection: AgentSquare [89] jointly searches P_i, M_i, W_i, and $\mathcal{D}$; OMAC [90] iteratively refines configurations, practically focusing on P_i; AgentInit [91] targets $R_i \times \mathcal{D}$ and composition by generating and selecting experts for relevance and diversity; and TUMIX-Evolve [93] optimizes diversity and utility by generating agents for tool-use reasoning. Conversely, Artemis [92] applies evolutionary search to full agent configurations (P_i, W_i, models) to balance cost and performance. Finally, MAEL [94] shifts focus to private memory M_i, leveraging cross-task experience pooling and retrieval to enhance collaboration.

At the system level, offline search overwhelmingly targets topology Γ, differing primarily in system representation. A prominent direction searches directly in **code space**. Pioneering this, ADAS [12] iteratively refines coded workflows via a meta-agent. AFlow [13] explicitly formalizes this code-level search, replacing ADAS's inefficient linear heuristics with a more effective procedure to discover workflows under budget constraints. A²Flow [97] further improves synthesis generalization by extracting reusable abstract operators to search within a more compact operator-level code space. Other methods instantiate similar code-based patterns with different emphases: Yuksel *et al.* [95] iteratively modify code over Γ, R_i, Λ, and $\mathcal{D}$; Nexus Architect [96] highlights failure-driven prompt refinement; JudgeFlow [98] applies block-level updates; and SCALE [99] boosts search efficiency by replacing execution with calibrated self-prediction. Meanwhile, other methods replace strict executable code with **natural language (NL) programs**. AutoFlow-C (closed-source AutoFlow [100]) iteratively refines NL workflows via an LLM generator and execution feedback. SwarmAgentic [101] uses language-driven population search to construct agents and jointly optimize roles, policies, and collaboration. Compared to code space, this NL space flexibly covers a broader range of workflows, coordination

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patterns, and role assignments without syntax constraints, though its weaker structural constraints complicate evaluation and refinement.

Another group outputs **graph-structured artifacts**. DebFlow [102] refines operator DAGs via multi-agent debate and execution feedback, whereas MermaidFlow [103] replaces debate with evolutionary search over compiler-verifiable flowcharts. AutoFlowGen [104] shifts to decomposition-driven synthesis, progressively building executable DAGs from natural language via reflection. Cognify [105] expands to directed workflow graphs, explicitly balancing quality, cost, and latency in its hierarchical search. Conversely, SOHM-D [106] solely targets communication topology, using RL-based search over inter-agent links. Distinctly, AgentXRay [107] uses pruned MCTS to reconstruct an explicit surrogate workflow graph rather than designing one from scratch.

Compared with the directly deployable methods above, another hybrid subset outputs specially designed artifacts but actively relies on them during test time. EvoFlow [22] uses evolutionary search to build a workflow library for online query-adaptive retrieval. ANN [108] optimizes a layered team graph via textual backpropagation for dynamic execution-time coordination. MetaAgent [110] generates a finite-state system whose searched states dictate test-time execution. ReSo [111] similarly governs online routing by combining task/agent graphs with reward-guided UCB subtask assignment. AgentNet [112] relies on evolved graphs for decentralized test-time routing and task forwarding. Conversely, MATTRL [113] constructs a shared pool of textual experiences from prior multi-agent interactions to support retrieval-augmented team formation and deliberation online.

Finally, some methods optimize components via **distinct search pipelines** rather than unified loops. GPTSwarm [114] separates RL-based topology (Γ) search from LLM-based prompt (P_i) updates. H-Swarms [115] decouples system-level communication DAG search from individual model weight adjustments. AgentDropout [109] factorizes topology optimization into node and edge dropout, learning agent participation and communication sparsity in two stages. MASS [116] explicitly stages optimization: warm-starting prompts, searching a pruned topology space, and then performing workflow-level prompt refinement on the selected structure.

3.3. Design-Time Adaptive MAS via Offline Policy Training

Offline policy training methods realize P4a by learning policies before deployment to task-conditionally instantiate or guide MAS configurations. Unlike model-training P4a methods, they learn explicit configuration policies rather than merely enhancing backbone capabilities. As Table 4 shows, they span multiple levels but are dominated by system-level, topology-centric policies, leaving agent-level and low-level underexplored. Fundamentally, they act as amortized optimizers: intensive offline learning at design time enables lightweight one-shot or iterative reuse at test time, naturally forming a **P4a+P4b hybrid**. Essentially, this learns a *query-to-configuration mapping*, though current methods typically cover only narrow configuration slices rather than the full space.

We organize the remainder primarily by *policy target component(s)*, threading from the dominant pure-topology policies to broader mixtures and non-topology groups. **Pure-topology methods** exclusively learn offline policies targeting the communication or workflow graph. DynaSwarm [131] and SOHM-L [106] train task-conditioned topology predictors, whereas AMAS [137] learns a per-query selector over RL-mined graphs. In specialized settings, CARD [141] optimizes condition-aware communication graphs for utility-cost tradeoffs, and DAWN [143] learns federated topology synthesis via structurally regularized aggregation. Differing mainly in training signals, RobustFlow [135] leverages instruction augmentation and preference optimization to learn query-conditioned workflows robust to semantic variations.

Beyond pure topology, another cluster augments **topology-led learning with agent-level components**. AutoFlow-O (open-source AutoFlow [100]) uniquely remains purely P4a, directly deploying offline-learned policies for topology and prompts. Generatively, ScoreFlow [127], FlowReasoner [128], and MAS-GPT [134] extend topology with prompts (and roles for the first two). AGP [129] and ARG-Designer [142] couple structure generation with agent composition, while MasRouter [130] and BAMAS [144] integrate model choice. ResMAS [145] stages offline topology learning, topology-aware

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prompt refinement, and target-task adaptation. Conversely, **purely agent-level policies** are rare: COPPER [118] and M2CL [119] shift offline learning entirely to prompts/instructions, ignoring system structure.

Table 4. Summarizing P4a-rooted offline policy training methods through policy characterization and usage. P4a and P4b refer to how the learned policy is used at design time and test time, respectively.

Level	Method	Policy Characterization: Target	Policy Characterization: Function in MAS	Policy Usage: P4a	Policy Usage: P4b
Agent Level	COPPER [118]	P_i	Generate role-specific reflections to revise task-agent prompts	✗	Iterative
Agent Level	M2CL [119]	P_i	Generate round-wise instructions to guide coherent consensus	✗	Iterative
System Level	*Mainly focus on scheduling policy Λ / task decomposition $\mathcal{D}$*				
	AOP [120]	$\Lambda, \mathcal{D}$	Generate and refine agent-oriented task plans	✗	Iterative
	Workforce [121]	$\mathcal{D}$	Generate domain-agnostic task decompositions	✗	Iterative
	Puppeteer [122]	Λ	Dynamically sequence and prioritize agent activations	✗	Iterative
	MPDF [123]	Λ	Adapt each agent's deliberation strategy from cognitive and peer states	✗	Iterative
	CD^{3}T [124]	$\Lambda, \mathcal{D}$	Decompose the task and assign agents to subtask-specific policies	✗	Iterative
	iMAD [125]	Λ	Escalate queries from single-agent reasoning to multi-agent debate	✗	One-shot
	BiRouter [24]	Λ	Route tasks to the next agent using query, history, and candidates	✗	Iterative
	AMRO-S [126]	Λ	Select semantic-aware execution paths	✗	Iterative
	Mainly focus on topology Γ				
	AutoFlow-O [100]	Γ, P_i	Generate executable natural-language workflows for a task type	One-shot	✗
	ScoreFlow [127]	Γ, R_i, P_i	Generate adaptive code-structured workflows for each task	✗	One-shot
	FlowReasoner [128]	Γ, R_i, P_i	Reason over execution feedback to generate one system per query	✗	One-shot
	AGP [129]	Γ, R_i	Predict a task-adaptive agent subset and communication graph	✗	One-shot
	MasRouter [130]	Γ, ϕ_i, R_i	Route collaboration mode, roles, and LLM backbones per query	✗	One-shot
	DynaSwarm [131]	Γ	Choose the best swarm communication graph for each query	✗	One-shot
	G-Designer [132]	Γ, C	Design the best communication graph for each query	✗	One-shot
	MaAS [133]	$\Gamma, P_i, W_i, \Lambda$	Sample a query-specific workflow from the agentic supernet	✗	Iterative
	MAS-GPT [134]	Γ, P_i	Generate a query-specific executable MAS in one inference	✗	One-shot
	RobustFlow [135]	Γ	Generate a stable query-specific workflow across synonymous queries	✗	One-shot
	SOHM-L [106]	Γ	Predict task-conditioned communication topologies for the swarm	✗	One-shot
	AdaptFlow [136]	$\Gamma, R_i, P_i, \Lambda$	Meta-learn a shared workflow initialization for fast subtask adaptation	✗	One-step
	AMAS [137]	Γ	Select the best communication graph for each query	✗	One-shot
	DyFlow [138]	$\Gamma, P_i, \Lambda, \mathcal{D}$	Plan and replan subgoal subgraphs from intermediate feedback	✗	Iterative
	EvoRL [139]	Γ, ϕ_i, Λ	Build a query-specific workflow and choose cost-aware LLMs for it	✗	Iterative
	MAS^2 [140]	$\Gamma, \phi_i, R_i, W_i, C$	Generate, implement, and rectify a task-specific MAS	✗	Iterative
	CARD [141]	Γ	Generate a condition-aware communication graph for each query	✗	One-shot
	ARG-Designer [142]	Γ, R_i	Assemble a task-specific agent graph from scratch	✗	One-shot
	DAWN [143]	Γ	Generate a federated, task-specific communication graph for each query	✗	One-shot
	BAMAS [144]	Γ, ϕ_i	Provision a budget-feasible LLM pool and choose collaboration topology	✗	One-shot
	ResMAS [145]	Γ, P_i	Generate a resilient MAS topology and refine topology-aware prompts	Adaptation	One-shot
	FlowSteer [146]	Γ, P_i, Λ	Interactively build and repair a workflow through canvas feedback	✗	Iterative
Low Level	ECON [147]	C	Learn a belief-based protocol replacing explicit agent communication	✗	One-shot
Low Level	ACL-LFT [148]	$\mathcal{K}$	Adaptively control the shared historical state/context passed to agents	✗	Iterative
Low Level	C2C [149]	C	Replace text communication with direct KV-cache communication	✗	One-shot
Low Level	VIL2C [150]	C	Prioritize high-VoI messages and stop reception early	✗	Iterative

Another group **extends topology-led learning to additional system-level components**. MaAS [133] learns an offline controller to jointly optimize topology, scheduling, and tool choice. AdaptFlow [136] meta-learns a reusable workflow initialization encoding scheduling structure. DyFlow [138] couples topology with scheduling and explicit task decomposition via learned subgoal-subgraph planning. EvoRL [139] combines topology and scheduling via hierarchical DQN, while FlowSteer [146] uses end-to-end RL over workflow edits and control structures. Crucially, these policies determine not just *which graph to use*, but *how it is organized and executed* system-wide.

Remaining **system-level policies target scheduling and/or task decomposition**. Scheduling-oriented methods including MPDF [123], iMAD [125], BiRouter [24], AMRO-S [126], and Puppeteer [122] train policies for deliberation control, escalation, routing, or RL-based orchestration. Conversely, Workforce [121] solely learns a reusable decomposition policy via filtered trajectories and preference optimization. AOP [120] and CD^{3}T [124] couple both Λ and $\mathcal{D}$: CD^{3}T learns subtask-specific decomposition-and-assignment, while AOP learns agent-oriented planning with evaluation and sub-task modification. Notably, AOP continuously updates its policy via a persistent cross-query feedback loop, making it a rare P4a+P4c hybrid.

Finally, a small group **extends topology-led learning to the communication layer**. G-Designer [132] augments offline graph design with communication-protocol control to organize inter-agent interactions. MAS[2] [140] broadens this by jointly learning policies across topology, roles, communication protocols, tools, and model choice. **Pure low-level policies** are even rarer. ECON [147], C2C [149], and VIL2C [150] target communication protocols: ECON learns an implicit belief-based protocol, C2C replaces text with direct KV-cache communication, and VIL2C learns message prioritization and early reception stopping. Conversely, ACL-LFT [148] uniquely targets cache/state sharing, learning to adaptively control shared historical context rather than redesigning the protocol itself.

4. Test-Time Adaptive Multi-Agent Systems

Although Sections 3.2 and 3.3 already covered several methods with test-time adaptation, those methods remain primarily centered on design-time optimization: they first search for a reusable artifact offline and then apply it at inference time, and are thus better viewed as P4a+P4b hybrids. Here we instead focus on methods whose center of gravity lies in P4b itself. By optimizing MAS configurations directly during inference, P4b methods offer stronger instance-level adaptability than P4a offline search, enabling dynamic query-specific tailoring and runtime anomaly response. However, because such optimization typically lacks deterministic ground truth and reliable verifiers, and therefore depends on softer feedback, it is also budget-constrained, feedback-limited, and less stable. As summarized in Table 5, the landscape is still dominated by system-level methods, most of which mainly optimize topology Γ, often together with closely coupled components such as Λ, R_i, P_i, or $\mathcal{D}$. Moreover, beyond utility, many of these methods place particular emphasis on efficiency. At the same time, relatively reliable feedback signals remain rare.

Table 5. Summarizing P4b methods through optimization-centric characterization. LL stands for low-level.

	Method	Optimization-Centric Characterization			
		Space / Target	Paradigm	Feedback Signal†	Obj. Beyond Utility
Agent Level	EvoAgent [151]	$R_i \times P_i$	Evolutionary	◐ LLM Evaluator	Diversity
	E3-MAS [152]	P_i	LLM-as-Optimizer	◐ LLM Evaluator, Execution Feedback	Efficiency, Reliability
System Level *Mainly focus on topology Γ*	AgentVerse [153]	$\Gamma \times R_i \times P_i \times \mathcal{D}$	LLM-as-Optimizer	◐ LLM Evaluator, Execution Feedback	Efficiency
	MorphAgent [154]	$\Gamma \times R_i$	Reflection-Based	◐ Execution Feedback, Self-Reflection	Robustness
	DyLAN [155]	$\Gamma \times R_i \times \Lambda$	Search-Based	◐ LLM Evaluator	Efficiency
	Flow [156]	$\Gamma \times R_i \times \Lambda \times \mathcal{D}$	Search-Based	◐ LLM Evaluator, Execution Feedback	Efficiency, Modularity
	IoA [157]	$\Gamma \times \Lambda \times \mathcal{D} \times \mathcal{C}$	LLM-as-Optimizer	○ Execution Feedback	Flexibility, Scalability
	AgentPrune [158]	Γ	Gradient-Based	● Task-Specific Evaluator	Efficiency, Robustness
	HALO [159]	$\Gamma \times R_i \times P_i \times \Lambda \times \mathcal{D}$	Search-Based	◐ LLM Evaluator	—
	MegaAgent [160]	$\Gamma \times R_i \times P_i \times \Lambda \times \mathcal{D}$	LLM-as-Optimizer	◐ LLM Evaluator, Execution Feedback	Reliability, Scalability
	Polymath [161]	$\Gamma \times \mathcal{D}$	Search-Based	◐ LLM Evaluator, Execution Feedback	Efficiency
	SwarmSys [162]	$\Gamma \times \Lambda$	Search-Based	◐ LLM Evaluator, Execution Feedback	Robustness, Scalability
	GoA (Taejong *et al.*) [163]	Γ	Search-Based	◐ Embedding Evaluator	Efficiency
	GUARDIAN [164]	Γ	Gradient-Based	◐ Agent Outputs, Communication Traces	Efficiency, Safety
	MAS-ZERO [165]	$\Gamma \times P_i \times \mathcal{D}$	LLM-as-Optimizer	◐ LLM Evaluator, Execution Feedback	—
	GoA (Yun *et al.*) [166]	Γ	Rule-Based	◐ Agent Outputs, LLM Evaluator	Efficiency, Scalability
	SelfOrg [167]	$\Gamma \times \Lambda$	Rule-Based	◐ Agent Outputs, Embedding Evaluator	Efficiency, Robustness
	SafeSieve [168]	Γ	Rule-Based	● LLM Evaluator, Task-Specific Evaluator	Efficiency, Robustness
	MASFly [169]	$\Gamma \times P_i \times \Lambda$	LLM-as-Optimizer	◐ LLM Evaluator, Execution Feedback	Robustness
	AdaptOrch [170]	$\Gamma \times \Lambda$	Rule-Based	◐ Embedding Evaluator, LLM Evaluator	Efficiency
Others	AutoAgents [171]	$R_i \times \Lambda \times \mathcal{D}$	LLM-as-Optimizer	◐ LLM Evaluator, Execution Feedback	—
	Captain Agent [172]	$R_i \times \Lambda \times \mathcal{D}$	LLM-as-Optimizer	◐ LLM Evaluator, Execution Feedback	Diversity
	SYMPHONY [173]	$\mathcal{M}_{\text{shared}} \times \Lambda$	Search-Based	◐ LLM Evaluator, Execution Feedback	Efficiency, Robustness
	BCCS [174]	$\Lambda \times \mathcal{J}$	Rule-Based	◐ Agent Outputs, Belief Scores	Robustness
LL	ACE [175]	$\Lambda \times \mathcal{C}$	LLM-as-Optimizer	◐ Agent Outputs, LLM Evaluator	—

[†] The leading mark denotes feedback signal reliability. ●: high, ◐: low, and ○: not applicable (no explicit feedback signal supporting runtime optimization).

At the agent level, P4b is limited: EvoAgent [151] generates agents via inference-time evolutionary search, while E3-MAS [152] iteratively refines faulty prompts through an execution–evaluation–evolution loop. The bulk of P4b is system-level, heavily dominated by topology-centric methods. A first cluster targets **pure topology**. In an amusing case of academic naming collision, two independent methods share the "GoA" moniker: Taejong *et al.* [163] search for input-dependent collaboration graphs for long-context compression, whereas Yun *et al.* [166] build query-specific communication graphs via relevance-aware node and edge selection. Targeting safety, GUARDIAN [164] detects anomalies in temporal interaction graphs to prune harmful links at runtime. Relying on stronger task-linked feedback, AgentPrune [158] performs one-shot pruning on sparse spatial-temporal structures based on utility, while SafeSieve [168] progressively prunes graphs using semantic initialization, edge contribution, and structure-aware clustering.

Topology-centered P4b methods often **co-optimize other components**. MorphAgent [154] rewrites agent profiles via execution feedback and self-reflection, while Polymath [161] jointly refines task-flow graphs and hierarchical decompositions. Broadening topology toward coordination, SwarmSys [162] uses event-driven matching and pheromone-like reinforcement, SelfOrg [167] builds communication DAGs directly from current responses, and AdaptOrch [170] routes decomposed tasks to various topologies. This trend deepens in broader system reconfiguration. AgentVerse [153] and DyLAN [155], two early and widely cited examples, adapt runtime team composition—the former via iterative recruitment, the latter via importance-based deactivation. Flow [156] dynamically updates workflows, assignments, and plans, emphasizing modular orchestration. Despite sharing similar configuration spaces, HALO [159] uses hierarchical refinement and MCTS subtask search, whereas MegaAgent [160] employs recursive recruitment and monitored queue-based coordination. Extending this, MAS-ZERO [165] iteratively revises sub-MASs from building blocks, while MASFly [169] instantiates query-specific systems from retrieved SOPs under active supervision.

A distinct line emphasizes **communication mechanisms**. IoA [157] heavily weighs instant-messaging-style coordination and LLM-driven FSMs for mixed synchronous/asynchronous interactions. This naturally leads to ACE [175], the only explicitly **low-level** method here, which adapts debate protocols online by dynamically composing next-round discussion acts. Outside the topology-centered majority, several methods target **other system-level coordination components**. AutoAgents [171] and Captain Agent [172] dynamically organize runtime teams and task flows: the former focuses on planner–observer drafting, while the latter emphasizes step-wise decomposition and expert retrieval. Distinctively, SYMPHONY [173] introduces shared memory via pool-wise reflection during planning. Lastly, BCCS [174] stands out as one of the few methods that explicitly optimize the collective decision mechanism $\mathcal{J}$ via belief-calibrated consensus and adaptive role assignments.

5. Self-Evolving Multi-Agent Systems

Unlike previous one-shot adjustments, P4c introduces cross-task persistent write-back: rather than optimizing transiently within the current instance, the system preserves acquired experiences, structures, or capabilities for future tasks. Representing the most advanced frontier of P4, this sustained self-evolution accumulates historical outcomes across task streams to alleviate cold-start costs, enhance adaptability, and enable continual improvement. However, it risks inheriting and solidifying low-quality or locally effective heuristics. Broadly, this evolution exhibits two forms: **Incrementally Persistent** optimization continuously updates existing components within a fixed configuration space, whereas **Expansively Persistent** optimization persistently expands the space of adjustable components itself, extending to objects like roles, tools, and topology. As Table 6 shows, the former dominates current literature, typically utilizing private or shared memory as a persistent substrate, while the latter targets more explicitly bounded configuration objects. Overall, although this line most clearly reflects the transition from transient to continual optimization, it remains relatively underexplored. Most existing methods endow only a small subset of components with persistent self-evolution, rather than making the entire MAS fully self-evolving. Moreover, strong guardrails and cleanup mechanisms remain underdeveloped, leaving systems vulnerable to error accumulation, unsafe capability carryover, and long-term entrenchment of flawed updates.

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Table 6. Summarizing P4c methods through persistence-centric characterization.

	Method	Optimization Components[†]	Persistence Mechanisms: Trigger[⋆]	Cleanup[⋄]	Guardrail[§]	Feedback Signal[‡]
Incrementally Persistent	AOP [120]	M_i, Λ, $\mathcal{D}$	SG	DD	SG	◐ LLM Evaluator, Task-Specific Evaluator
	SEDM [176]	$\mathcal{M}_{\text{shared}}$	ED	DM, MG, PR	TR, VG	● Execution Feedback, Task-Specific Evaluator
	AgentEvolver [65]	ϕ_i, $\mathcal{M}_{\text{shared}}$	ExD	—	VG	◐ Execution Feedback, LLM Evaluator
	G-Memory [177]	$\mathcal{M}_{\text{shared}}$	ExD	—	—	◐ Execution Feedback, LLM Evaluator
	AgentNet [112]	Γ, M_i, Λ	ExD	PR	—	◐ Execution Feedback
	BiRouter [24]	$\mathcal{M}_{\text{shared}}$, Λ	ED	DM	VG	◐ Execution Feedback, LLM Evaluator
	Wang *et al.* [178]	$\mathcal{M}_{\text{shared}}$, Λ	SG	—	SG	◐ Entropy-Based and Task-Specific Evaluator
	Agora [179]	$\mathcal{M}_{\text{shared}}$, C	CFD	—	FE, VG	● Communication Frequency, Execution Feedback
Expansively Persistent	AutoAgent [180]	Γ, R_i, W_i	ErD, RD	—	CB, ET	● Constraint Feedback, Execution Feedback
	AutoMaAS [181]	Γ, Λ	FD	PR	CB	◐ System Feedback, User Feedback
	AgentOrchestra [182]	P_i, M_i, W_i	ErD, FD	—	ET, RE	◐ Execution Feedback
	MetaGen [183]	Γ, R_i, P_i, Λ	FD, SG	—	CB, SG	◐ Execution Feedback

[†] **Black** for P4a, **Blue** for P4b, **Orange** for incrementally persistent components, and **Green** for expansively persistent components. [‡] The leading mark denotes feedback signal reliability. ●: high, ◐: low. [⋆] What event or signal initiates persistent updates. SG: Success-Gated; ED: Evaluation-Driven; ExD: Experience-Driven; CFD: Communication-Frequency-Driven; ErD: Error-Driven; RD: Requirement-Driven; FD: Feedback-Driven. [⋄] Whether and how persistent content is removed, merged, or down-weighted over time. DD: Deduplication; DM: Demotion; MG: Merge; PR: Pruning. [§] What mechanism prevents low-quality persistent updates from being solidified. SG: Success-Gated; VG: Verifier-Gated; TR: Traceable; FE: Fallback-Enabled; CB: Constraint-Based; ET: Execution-Tested; RE: Rollback-Enabled.

Memory-centric incremental persistence forms a major P4c branch, writing execution outcomes into reusable substrates. While some appeared in Section 3 regarding design-time construction, we now focus on their test-time persistence. SEDM [176], AgentEvolver [65], G-Memory [177], BiRouter [24], and Wang *et al.* [178] center on $\mathcal{M}_{\text{shared}}$, differing in persisted content and control mechanisms. SEDM maintains a repository via verified admission and features explicit cleanup, acting as a rare method with strong guardrails. While both accumulate execution traces or insights, AgentEvolver requires explicit validation before integration, whereas G-Memory emphasizes hierarchical cross-task reuse. BiRouter persists cross-task reputation scores rather than raw trajectories, updating a shared estimate of agent reliability to shape routing. Wang *et al.* employs a success-gated mechanism, conservatively writing back only after successful collaborations. Agora [179] extends this by persisting both $\mathcal{M}_{\text{shared}}$ and C, consolidating frequently repeated interactions into new protocol conventions, yet crucially remains incremental by refining assets within an existing protocol substrate rather than enlarging the configuration space itself. Conversely, AOP [120] and AgentNet [112] rely on M_i. AOP stores representative prior works after successful task completion, using deduplication for control. AgentNet persistently updates both M_i with agent-specific trajectories and knowledge and Γ via task-driven edge-weight updates and pruning, keeping this topology adaptation incremental by merely adjusting connectivity within a fixed graph rather than expanding the structural search space. Overall, memory-centric approaches dominate current P4c research because memory is the most natural and easy-to-operationalize form of persistence: it supports cross-task accumulation without requiring full system reorganization after each task.

Expansively persistent optimization shifts persistence from accumulated memory-like substrates to the growth of explicitly bounded configuration objects. AutoAgent [180] is the clearest pure case, persistently expanding Γ, R_i, and W_i through requirement-driven construction and error-triggered regeneration. The remaining methods combine expansive and incremental persistence. AutoMaAS [181] expansively evolves Γ while incrementally updating Λ, so structural growth is accompanied by continued refinement of scheduling decisions. AgentOrchestra [182] persistently expands W_i, while incrementally updating P_i and M_i through execution feedback, re-execution testing, and rollback support. MetaGen [183] similarly expands R_i by generating and solidifying new roles, while incrementally carrying forward updates to P_i and Λ through feedback- and success-gated write-back. Compared with memory-centric methods, these approaches make persistence more explicitly visible at the level of system organization and capability growth. At present, expansive persistence remains concentrated on a small set of components, especially roles, tools, and topology. This likely reflects the current stage of exploration rather than an intrinsic boundary, since these are simply the components most readily externalized and expanded, not the only ones that could support expansive persistence.

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6. Automated Multi-Agent Systems for Domain-Specific Applications

Unlike the general-purpose methods discussed in previous sections, a substantial body of work targets domain-specific settings as shown in Table 7, revealing how domain constraints reshape *which* components are optimized and *why*, rather than introducing entirely new paradigms. While these methods span all P4 sub-paradigms, their distribution is highly skewed, heavily dominated by healthcare and software engineering (SE). Different domains naturally induce distinct optimization hotspots: healthcare drives optimization across all sub-paradigms, especially persistent memory for long-term clinical expertise, while SE targets topology, prompts, and scheduling to counter long-horizon execution brittleness. Furthermore, hard operational constraints favor explicit system-level optimization, with embodied systems prioritizing coordination for safety, finance and enterprise focusing on topology and memory for risk control, and scientific problem-solving relying on tools and decomposition to manage structured dependencies. Across these domains, persistent self-evolution concentrates where experience naturally accumulates, yet typically remains localized to specific components like memory or prompts rather than expanding to the full MAS. Ultimately, these methods are best viewed as *domain-shaped instantiations* of general P4 principles, while also serving as stress tests that expose limitations in feedback reliability, safety, and long-term evolution under real-world constraints.

Table 7. Summarizing domain-specific Automated MAS methods. Opt. stands for optimization.

	Method	P4a	P4b	P4c	Opt. Target[†]	Domain-Specific Bottleneck
Healthcare	MedAgent-Zero [184]			✓	M_i	Continual clinical expertise accumulation from simulated practice
	MDTeamGPT [185]			✓	$\mathcal{M}_{\text{shared}}$	Reusable diagnostic knowledge under long MDT deliberations
	SeM-Agents [186]			✓	$\mathcal{M}_{\text{shared}}$	Evolving consultation experience beyond static zero-shot medical teams
	EvoPatient [187]			✓	$\mathcal{M}_{\text{shared}}$	Learning realistic yet medically grounded patient presentation patterns
	STELLA [188]			✓	$\mathcal{M}_{\text{shared}}$, W_i	Evolving biomedical tool use amid fragmented databases, tools, and literature
	IAAG [189]		✓		Γ, R_i	One-shot expert expansion for context-specific medical consultation
	DRTAG [189]		✓		Γ, R_i, Λ	Continual expert expansion as diagnostic needs shift during dialogue
	MedAgentSim [190]			✓	$\mathcal{M}_{\text{shared}}$	Experiential improvement in interactive diagnosis with explicit testing
	Zhuang *et al.* [191]	✓			Γ, Λ, $\mathcal{D}$	Automated workflow architecture search for diverse clinical scenarios
	MedDCR [192]	✓			Γ, Λ, $\mathcal{D}$	Medically valid coding workflows under rules and tools
	Warrier *et al.* [193]	✓	✓		Γ, Λ, $\mathcal{D}$	Distributed orchestration of volatile multi-department healthcare workflows
	MMedAgent-RL [194]	✓			Γ, Λ, $\mathcal{J}$	Optimized specialist routing and aggregation across medical subdomains
Software Engineering	Co-Learning [195]	✓		✓	M_i	Cross-task reuse of collaboration experience in multi-step development
	ToP [196]			✓	Γ, $\mathcal{M}_{\text{shared}}$, Λ, $\mathcal{D}$	Reusable and hallucination-resistant software process generation
	EvoMAC [197]		✓		Γ, R_i, P_i	Task-adaptive coding collaboration beyond static human-authored workflows
	SEW [198]	✓			Γ, P_i	Automatic workflow and prompt design for diverse coding tasks
	Shen *et al.* [199]		✓		P_i	Targeted prompt repair in role-based software collaboration
	EvoConfig [200]			✓	P_i, $\mathcal{K}$	Continual diagnosis and repair of environment configuration failures
	AgentSpawn [201]		✓		Γ, R_i, Λ, $\mathcal{K}$	Runtime expansion for long-horizon code tasks discovered on the fly
	MARS^2 [202]	✓			Γ, ϕ_i, R_i, Λ	Heterogeneous search-and-learn beyond single-agent coding ceilings
	VFlow [203]	✓			Γ, Λ, $\mathcal{D}$	Multi-objective workflow search under hardware verification constraints
Finance	FINCON [204]	✓			Γ, P_i	Long-horizon risk-aware financial decision making from multi-source signals
	SOAN [205]			✓	Γ, R_i	Structure-aware self-organization for deep and nested enterprise workflows
	HiveMind [206]			✓	P_i	Continual prompt repair under shifting financial environments
	LEGOMem [207]	✓			M_i, $\mathcal{M}_{\text{shared}}$	Reusable multi-agent procedural memory for enterprise task execution
Law	AgentCourt [208]			✓	$\mathcal{M}_{\text{shared}}$	Evolving legal reasoning for dynamic adversarial courtroom interaction
Embodied Systems	HAS [209]		✓		Γ, M_i, Λ	Dynamic multimodal coordination in open-world embodied environments
	CaPo [210]		✓		Λ, $\mathcal{D}$	Long-horizon strategic cooperation under partial observability
	RALLY [211]	✓	✓		R_i, $\mathcal{J}$	Interpretable heterogeneous coordination for UAV swarms
	HMARL-CBF [212]	✓			ϕ_i, Λ	Scalable safe coordination with pointwise safety guarantees
	CAML [213]	✓			ϕ_i, $\mathcal{C}$	Robust driving collaboration under missing test-time modalities
	RMTC [214]	✓			Γ, R_i, Λ	Heterogeneous temporal coordination of lights and emergency vehicles
	Kawabe *et al.* [215]			✓	P_i, Λ, $\mathcal{D}$	Planner-grounded prompt evolution for long-horizon robot collaboration
Information Seeking	AgentCF [216]	✓			M_i	Two-sided user–item interaction modeling beyond text-only histories
	Mobile-Agent-E [217]			✓	$\mathcal{M}_{\text{shared}}$, W_i, Λ	Long-horizon multi-app exploration with reusable mobile skills
	SMART [218]	✓			ϕ_i	Distractor-robust knowledge use under fixed multi-agent RAG structure
	FlowSearcher [219]			✓	Γ, $\mathcal{M}_{\text{shared}}$, $\mathcal{D}$	Long-horizon branching web research with reusable structural experience
	DualAgent-Rec [220]		✓		Λ, $\mathcal{J}$	Constrained recommendation under hard fairness and exposure requirements
Scientific Problem Solving	HeurAgenix [221]	✓			R_i, Λ, $\mathcal{J}$	Adaptive end-to-end heuristic discovery for new optimization problems
	ComfyGPT [222]	✓			Γ, W_i, $\mathcal{D}$	Hallucination-resistant generation of complex ComfyUI workflows
	SIER [223]		✓		R_i, $\mathcal{J}$	Maintaining solution diversity in multi-agent mathematical reasoning
	EcoLANG [224]	✓			$\mathcal{C}$	Token-efficient yet intention-preserving communication in social simulation
	GeoFlow [225]		✓		Γ, W_i, $\mathcal{D}$	Error-correctable GIS workflow orchestration across multiple APIs
	MATA [226]	✓	✓		$\mathcal{M}_{\text{shared}}$, Λ	Learned switching among reasoning modes in compositional visual tasks
	SEVADE [227]		✓		Γ, R_i, Λ, $\mathcal{J}$	Dynamic viewpoint diversification for robust sarcasm detection

[†] **Orange** for incrementally persistent components, and **Green** for expansively persistent components.

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7. Toward Agentic Swarms: Early Explorations and Outlook

While P4 methods transcend static orchestration, they still optimize within bounded configuration spaces using explicitly representable scaffolds. This becomes inadequate in open-ended, large-scale environments where collective capability must arise from interaction rather than the refinement of predefined templates. Moving beyond structurally bounded coordination toward decentralized interaction, emergent differentiation, and scale-sensitive dynamics is the core motivation for P5, formally defined as Agentic Swarms in Definition 13. Since a mature P5 paradigm has not yet emerged, we use Table 8 not as an absolute boundary test, but as a structured lens to examine how existing works probe this regime along three core dimensions: decentralization, emergence evidence, and scale sensitivity. Overall, methods strongly exhibiting all three dimensions remain rare, with most current efforts advancing only one or two aspects.

Table 8. Extent to which methods explore the core dimensions of Agentic Swarms. Dec., Emer., and Scale denote Decentralization, Emergence Evidence, and Scale Sensitivity, respectively. Note that the methods in the gray-shaded rows are technically outside the scope of P4.

Method	Dec.[†]	Emer.[†]	Scale[†]	P5 Exploration	Method	Dec.[†]	Emer.[†]	Scale[†]	P5 Exploration
SwarmSys [162]	◐	◐	◐	★★★	AgentNet [112]	●	◐	◔	★★★⯪
DAWN [143]	◔	○	◔	★	BiRouter [24]	●	◐	○	★★★
SelfOrg [167]	◐	◐	◐	★★★	MARSHAL [53]	○	◐	○	★
ExpoComm [77]	◐	○	◐	★★	COALA-PG [45]	●	◐	○	★★★
MAPoRL [35]	○	◐	○	★	PARCO [43]	○	○	◐	★
RAC [50]	●	○	○	★★	TRPE [62]	●	○	○	★★
DR-AC [59]	●	○	○	★★	FlickerFusion [67]	○	○	◐	★
DoF [68]	◔	○	◐	★⯪	STAIRS-Former [73]	○	○	◐	★
CoMAS [39]	◐	◐	◐	★★★	MPDF [123]	◐	○	○	★
MorphAgent [154]	●	◐	◔	★★★⯪	IoA [157]	◔	◔	○	★
Agora [179]	●	●	◐	★★★★★	AgentVerse [153]	○	●	◔	★★⯪
H-Swarms [115]	○	◐	◐	★★	S-Agents [228]	◔	◐	◔	★★
PIANO [30]	●	●	●	★★★★★★	MacNet [229]	○	●	●	★★★★
Zhang *et al.* [230]	○	●	◔	★★⯪	Jimenez-Romero *et al.* [231]	◐	●	○	★★★
TDMI [232]	◐	●	◔	★★★⯪	CompeteAI [233]	◐	●	○	★★★
AgentSociety [234]	●	●	●	★★★★★★	Generative Agents [235]	◐	●	○	★★★

[†] ○: None (no meaningful exploration); ◔: Little (explicit but limited); ◐: Partial (substantial but still bounded); ●: Strong (central and well-supported).

At the lower end of this exploratory landscape, methods weakly probe the P5 regime along single dimensions, inheriting decentralization from MARL or relating to scale largely in the sense of scalability or robustness. A substantive middle group advances more unevenly. For instance, SwarmSys [162] balances all three dimensions via self-organizing explorer–worker–validator cycles and scaling studies, though its logic remains bounded. AgentNet [112] strongly emphasizes decentralization through distributed routing and adaptive graphs, but weakly explores emergence and scale. Conversely, TDMI [232] focuses primarily on emergence, providing an information-theoretic framework to diagnose role differentiation and collective transitions, with limited scale engagement. MorphAgent [154] bridges decentralization and emergence via self-evolving profiles and dynamic role adaptation, though its scale exploration remains modest.

At the strongest end, MacNet [229], Agora [179], and notably AgentSociety [234] and PIANO [30] offer the deepest P5 explorations. MacNet tackles emergence and scale by framing collaboration as a scaling object and revealing a logistic collaborative scaling law, though its coordination remains orchestrated via a DAG rather than fully decentralized. Conversely, Agora achieves strong decentralization through a central-authority-free protocol layer, studying the emergence of self-organizing communication within large agent networks. Providing the clearest glimpses of the full P5 regime, AgentSociety and PIANO provide the clearest glimpses of the full P5 regime by concurrently engaging all three dimensions. By pushing simulations to unprecedented societal scales, AgentSociety leverages over ten thousand agents to surface macro-level collective phenomena, such as information diffusion and economic behaviors, that are fundamentally unobservable in small-scale setups. Similarly, PIANO utilizes thousands of autonomous agents in open-ended environments to explicitly benchmark emergent civilizational dynamics, including role specialization and cultural transmission. Crucially, these

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works treat scale not merely as an auxiliary stress test, but as a fundamental constitutive variable for generating true collective emergence.

Collectively, Table 8 suggests that current literature primarily decomposes the prerequisites of Agentic Swarms rather than delivering a mature P5 paradigm. Progress remains fragmented. The critical next step toward true P5 is to jointly realize and measure these dimensions. Rather than treating scale merely as an efficiency stress test, the field must learn to harness it as a fundamental engine for new organizational capabilities. How to methodologically bridge this gap and how to design evaluation protocols capable of isolating true collective emergence remain defining challenges. We explore these issues as key future directions in Section 10.4.

8. Beyond Task Enumeration: Benchmarking Automated MAS Adaptation

While existing surveys have extensively reviewed MAS benchmark taxonomies [236–238], this section moves beyond cataloging static task sets to investigate a core question: *for automated MAS, what evaluation paradigm can pierce surface-level task completion to capture the essence of coordination adaptation*? Recent evolution reveals a shift from mere task accumulation toward the explicit operationalization of multi-agent challenges. By internalizing mechanisms like information asymmetry or long-term resource games, these benchmarks compel systems to exhibit collaboration transcending single-agent reasoning. For instance, the goal-driven social interactions constructed in SOTOPIA and the resource governance pressures introduced by GOVSIM both exemplify this design philosophy, embedding multi-agent interaction logic as core environmental constraints [239,240].

This evolutionary logic can be deconstructed into two progressive levels. The primary level entails constructing a "collaboration-aware" evaluation framework. Its core shifts the metric focus from "task outcomes" to the "collaboration process," explicitly examining how inter-agent communication and coordination substantively contribute to final results. For instance, MultiAgentBench [241] integrates collaboration and planning into a unified framework; InformativeBench [242] operationalizes collaborative efficacy as the exchange efficiency of distributed private information; and Collab-Overcooked [243] leverages process metrics to fine-grainedly measure the differences between "initiating" and "responding to" collaboration. Thus, the defining criterion of a true MAS benchmark is whether collaborative behavior enters a quantifiable evaluation system, elevating it from a mere "multi-agent environment" to an "instrument for measuring collaborative efficacy."

Despite being collaboration-aware, current evaluations fail to penetrate the P4 paradigm's essence. The critical criterion is *discriminative power for automated adaptation*: P4-effective benchmarks must make performance highly sensitive to *dynamic configuration adjustment*. If fixed templates suffice, powerful base models use single-agent reasoning as a shortcut to mask architectural rigidity, capturing only coordination execution rather than adaptation. The core test is whether a system suffers performance decay or cost spikes due to its failure to reconstruct collaboration mechanisms. By this standard, benchmarks like MultiAgentBench [241], SwarmBench [244], GOVSIM [240], and MindAgent [245] mostly reside at the execution or behavioral policy layers, failing to trigger system-level reconfiguration. In contrast, evaluations of workflow synthesis or iterative refinement, such as Nexus Architect [96], more effectively test P4 capabilities by making process adjustment a decisive factor in performance.

Following this logic, evaluations must transcend single-metric calculations and elevate to a systematic framework: **anchoring the evaluation target, adopting specific protocols, and probing system behavior at defined granularities.** Existing explorations show that protocol design inherently dictates experimental conclusions. Benchmarks like WORFBENCH [246] and A2ASECBench [247] demonstrate that intermediate workflows and protocol interactions can be directly verified via target-aware matching and system-level probing, without relying solely on posterior task success. Furthermore, BattleAgentBench leverages communication ablation and pressure curves to enable failure localization and mechanism isolation [248], while MAST complements such top-down protocols with a bottom-up, failure-centric analysis of MAS breakdowns [249]. Evaluations are evolving toward framework-level isolation [250], surrogate evaluation for workflow search [251], and deployment-aware metrics like the "cost-of-pass" [252].

9. Safety in Automated Multi-Agent System Optimization

While the trustworthiness of LLM-based MAS has gained significant traction [253–256], existing research primarily targets static architectures and fixed protocols. These studies address critical risks: prompt injection propagating across agent boundaries, collective behavior corruption by compromised nodes, cascading failures triggered by local breakdowns, and privacy leakage within communication channels. Automated MAS inherits these threats, yet P4 introduces a unique "second attack surface": the optimization process $\mathcal{F}$ itself. As systems gain the autonomy to update their own configurations, safety concerns shift from transient runtime behaviors to the risk of optimization trajectories drifting into unsafe regions of the configuration space. We identify three novel risk categories: **optimization-induced safety degradation**, **persistent safety drift**, and **feedback poisoning**. The first stems from objective mismatch: when $\mathcal{F}$ prioritizes utility $U(\theta, \mathcal{T})$ over safety constraints $\mathcal{B}$, safety is traded for performance gains, embedding the resulting degradation directly into the optimized configuration θ^* as a structural vulnerability.

The second risk manifests through persistent cross-task optimization. Recent work shows that continuous self-evolution can produce unintended harm along four pathways: model, memory, tool, and workflow [25]. Specifically, self-training can erode alignment; memory can induce reward hacking; tool evolution may introduce vulnerabilities; and workflow optimization can amplify unsafe decisions. These risks are characterized by *persistence and propagation*: once unsafe elements are written back into the configuration, failures transcend single episodes to affect all future tasks. The third risk targets the feedback loop: since feedback $r = \mathcal{E}(\tau, g)$ is the core signal driving automated optimization, its adversarial manipulation can reshape the system's future configuration trajectory. Other trustworthiness dimensions, such as robustness, privacy, and fairness, remain largely unexplored in Automated MAS [257,258]; we defer these to Section 10.3.

10. Future Directions

10.1. Advancing the Optimization Foundations of Automated MAS

- **From Fragmented Tuning to Holistic Configuration Co-Optimization.** Current Automated MAS optimizations explore a sparse design space. Transient tuning disproportionately favors high-level topologies over micro-level dynamics, while persistent evolution remains memory-centric, leaving broader MAS assets largely unmapped. To break this fragmentation, the field need to transition toward holistic co-optimization frameworks. Future research should formulate the entire MAS configuration as a genuinely coupled continuum, developing algorithms capable of seamlessly co-adapting low-level protocols, macro-architectures, and persistent assets.
- **From Stage-Separated Tuning to a Lifecycle-Consistent Optimization Framework.** Current hybrid methods merely pipeline offline pre-training with online execution, rather than forming a cohesive evolutionary loop. The field must transition toward an integrated optimization lifecycle: offline learning should extract transferable MAS priors instead of fixed controllers; runtime control should execute efficient adaptation; and persistent write-back should selectively consolidate cross-task improvements. Crucially, a mature P4 must learn not only how to optimize, but also when *not* to—proactively restricting updates under ambiguous feedback or elevated risk.

10.2. Next-Generation Evaluation and Benchmarking

- **From Collaboration-Aware to Adaptation-Discriminative Benchmarks.** While existing evaluations capture static coordination, future frameworks should enforce *adaptation necessity*, actively penalizing architectural rigidity. Evaluations are expected to transition from episodic task-success rates to process-centric, long-horizon assessments. This requires explicitly quantifying continuous optimization trajectories, runtime reconfiguration agility, and the cumulative impact of

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persistent write-backs. Ultimately, systems should be judged not merely on executing predefined collaboration, but on efficiently evolving their configurations over time.

- **From Static Outcomes to Multi-Dimensional Lifecycle Profiling.** While current benchmarks prioritize snapshot task success, future frameworks should evaluate the entire automated MAS lifecycle. This requires longitudinal tracking to quantify dynamic configuration evolution over time. Concurrently, trustworthiness and safety should be elevated to primary evaluation targets to explicitly penalize mal-evolution. Furthermore, the optimization process $\mathcal{F}$ should be decoupled from its generated solutions and subjected to independent meta-evaluation for generalization and sample efficiency. Ultimately, as systems approach the open-ended interactions of P5, traditional target-aware probing should evolve into population-level behavioral analysis, providing scalable methodologies to scientifically measure macroscopic emergent phenomena.
- **From Sterile Verification to Justifying MAS Necessity.** While current methods thrive in sterile simulators, future frameworks should explicitly answer when multi-agent complexity is genuinely necessary. Bridging the research-to-deployment gap requires treating latency, token budgets, and coordination overhead as primary metrics to systematically justify MAS against lighter single-agent baselines. Furthermore, the messy realities of real-world deployment, including dynamic interfaces, noisy signals, and strict domain constraints like financial compliance or healthcare knowledge accumulation, should fundamentally reshape optimization theories. By anchoring evaluations in these practical friction points, the community can transition from merely proving that systems can collaborate to rigorously defining the exact boundary where the added coordination burden shifts from architectural overkill to a deployment-worthy necessity.

10.3. Trustworthiness for Automated MAS

- **From Post-Hoc Alignment to Inherently Safe Optimization Dynamics.** Building upon the three novel attack surfaces detailed in Section 9, future research could transition from diagnosing risks to engineering inherently safe solutions. Frameworks hold the potential to embed safety as a hard mathematical constraint within the optimization objective itself, actively preventing algorithms from trading alignment for utility. To mitigate persistent configuration drift, exploring automated rollback mechanisms and verifiable configuration versioning offers a promising avenue. Furthermore, securing feedback loop integrity calls for developing tamper resistant evaluation protocols that immunize evolutionary trajectories against adversarial poisoning. Ultimately, safety is expected to evolve from an external patch into an intrinsic architectural pillar governing automated self improvement.
- **From Safety to Holistic Trustworthiness in Automated Evolution.** Advancing beyond core safety, a broader trustworthiness framework should encompass dimensions uniquely reshaped by configuration optimization. This expansion involves several prospective dimensions that remain largely untapped in existing literature. *Robustness* transcends input stability to ensure architectural updates remain effective across task variations. Similarly, *reliability* in the presence of ambiguous feedback involves maintaining predictable optimization trajectories despite imperfect signals. *Privacy* extends into persistent state modifications and memory write-backs, while *fairness* evaluation examines whether optimization algorithms privilege specific architectural pathways. Finally, *explainability* holds the potential to demystify the evolutionary logic itself, clarifying why specific configurations are autonomously adopted. These dimensions represent critical research frontiers for securing the entire lifecycle of automated MAS deployment.

10.4. From P4 to P5: Advancing Toward Agentic Swarms

- **From Bounded Optimization to a Controllable Swarm Transition.** While P4 automates bounded configurations, transitioning to P5 demands more than mere scaling. The critical bridge lies in elevating P4c persistent assets from static states into the *institutional substrate* of artificial collectives. Future frameworks should formalize these persistent components as the underlying infrastructure through which emergent collective norms, role expectations, and interaction conventions

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dynamically stabilize over long horizons. The ultimate challenge is engineering a *controllable swarm transition*, steering bounded self-evolution toward decentralized, emergent organizations without relinquishing interpretability, evaluability, or systemic governance.

- **From Encodable Safety Constraints to Governing Emergent Phase Transitions.** P4 safety relies on bounding configuration spaces (e.g., restricting $\mathcal{B}$ or θ). This paradigm systematically collapses in P5 Agentic Swarms, where capabilities decouple from finite templates and emerge from open-ended interactions. Consequently, P5 systems face profound complex-system vulnerabilities: unpredictable phase transitions that bypass static rules, absolute accountability gaps driven by massive interactions, and uncontrollable cascading effects triggered by local interventions. Future research must urgently draw from statistical physics or ecology to pioneer frameworks capable of safely steering unbounded population dynamics.

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